SECTION 1: SYSTEM ARCHITECTURE & EXECUTIVE OVERVIEW

==========

Product Name: PSD PDF Studio and Viewer Enterprise Edition
(v3.0.0-ENTERPRISE-FLAGSHIP)
Vendor: PSD Software Studio / Prabhu Kannan
Licensing: MIT Open Source License (Permissive Enterprise & Commercial Usage)
Operating Guarantee: 100% Offline Air-Gapped Operation | Zero External Cloud Leakage

PSD PDF Studio and Viewer is a zero-dependency desktop productivity workstation engineered
to deliver ultra-fast vector PDF viewing, native OpenXML Word (.docx) editing, cryptographic
e-signatures, and local CPU-based neural document intelligence.

The architecture is strictly divided into two verified targets:
1. Native Windows 10/11 x64 Workstations, Ultrabooks, Surface Tablets, and Enterprise Servers.
2. Ultra-Lightweight Android Mobile Devices (specifically optimized for Qualcomm Snapdragon 625
architectures running Android 6.0 Marshmallow through Android 14+).

SECTION 2: WINDOWS DESKTOP & ENTERPRISE SERVER HARDWARE REQUIREMENTS

==========

2.1 MINIMUM SYSTEM HARDWARE (CLIENT WORKSTATION)

- Processor (CPU): 1.0 GHz 64-bit (x64) Intel Core 2 Duo / AMD Athlon 64 or newer.
- System RAM: 512 MB Physical RAM (Application strictly clamps working set under 150MB).
- Disk Storage: 160 MB available local disk space for full standalone binary and assets.
- Display Resolution: 1024 x 768 pixels with 16-bit color depth (High-DPI responsive).
- Operating Systems: Windows 10 Version 1903 (Build 18362) or newer (Windows 10, 11, x64).
- Runtime Dependencies: ZERO. Self-contained .NET 8 runtime, DirectX drivers, and WinRT APIs.

2.2 RECOMMENDED ENTERPRISE HARDWARE (HEAVY MULTI-PAGE BLUEPRINTS & CAD)

- Processor (CPU): Quad-Core Intel Core i5/i7/i9 (8th Gen+) or AMD Ryzen 5/7/9 (3.0 GHz+).

- System RAM: 2.0 GB Physical RAM or higher.
- Storage Device: 500 MB Solid State Drive (NVMe SSD recommended for instant library indexing).
- Graphics Hardware: DirectX 11 / Direct3D 12 compliant GPU for 120 FPS hardware rasterization.
- Display Resolution: 1920 x 1080 (Full HD) up to 3840 x 2160 (4K UHD) with 100%-250% DPI scaling.
- Enterprise Operating Systems: Windows 11 Enterprise/Pro, Windows 10 Enterprise LTSC,
  Windows Server 2019, Windows Server 2022, Windows Server 2025.

2.3 MEMORY CEILINGS & DIRECTX ACCELERATION BENCHMARKS

- 64MB LRU Tile Rasterizer Cache: Viewport tiles are cached in high-speed RAM with strict
  byte-eviction. Continuous scrolling across 1,000+ page documents will never cause memory growth.
- Native WinRT DirectWrite Acceleration: Native vector glyph rasterization runs directly on
  hardware, delivering fluid 60 to 120 FPS navigation with zero frame drops or visual tearing.

SECTION 3: ANDROID MOBILE HARDWARE REQUIREMENTS (QUALCOMM SNAPDRAGON 625)

==========

3.1 TARGET HARDWARE PROFILE

- System-on-Chip (SoC): Qualcomm MSM8953 Snapdragon 625 (14nm FinFET technology).
- CPU Core Architecture: Octa-core 2.0 GHz Cortex-A53.
- GPU: Qualcomm Adreno 506.
- Benchmark Reference Device: Xiaomi Redmi Note 4 (3GB RAM / 32GB Storage).

3.2 MOBILE OPERATING SYSTEM COMPATIBILITY

- Minimum API Level: Android 6.0 Marshmallow (API 23).
- Target API Level: Android 14.0 Upside Down Cake (API 34) and Android 15 (API 35).
- Binary ABI Architectures: Dual compiled native libraries:
  * armeabi-v7a (32-bit ARM with NEON SIMD vector extensions).
  * arm64-v8a (64-bit ARMv8-A architecture).

3.3 MOBILE HEAP CEILING & CAMERA SCANNING

- Dalvik / ART Heap Cap: strictly constrained to < 45 MB operational heap.
- Camera Scanner: Compatible with standard CMOS sensor modules for document scanning,

quad-corner perspective trapezoid correction, and high-contrast Otsu binarization.

SECTION 4: ENTERPRISE NETWORK, SECURITY & PACKAGING TELEMETRY

==========

- Network Connectivity: ZERO REQUIREMENT. All cryptographic operations, OCR layers, text converters,
and AI Copilot heuristics execute locally on the host CPU. No telemetry packets are transmitted.
- Cryptographic Standards: AES-256 (CBC mode) document permissions, SHA-256 integrity digests,
PKCS#7 detached digital signatures, and RFC 3161 compliant timestamp validation.
- Packaging Delivery Modes:
1. Standalone Executable: psd_pdf.exe (78.3 MB, portable double-click).
2. Enterprise Portable: PSD-PDF-Studio-Portable.zip (144.9 MB).
3. Inno Setup Client Installer: PSD-PDF-Studio-Setup.exe (143.6 MB).
4. WiX Enterprise MSI: PSD-PDF-Studio-Enterprise.msi (71.6 MB, msiexec /i /qn silent deployment).
5. Android Universal Release APK: PSD-PDF-Studio-Release-universal.apk (21.1 KB).